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(54) **PREVENTING DATA AND HARDWARE LOSS IN PRE-PROVISIONED DEVICES**

(71) Applicant: **Cisco Technology, Inc.**, San Jose, CA (US)

(72) Inventors: **William Sterling Alexander**, Durham, NC (US); **Wayne Elmer Moorefield, Jr.**, Midlothian, VA (US)

(73) Assignee: **CISCO TECHNOLOGY, INC.**, San Jose, CA (US)

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H04L 9/40 (2022.01)

(52) **U.S. Cl.**
CPC ***H04L 63/083*** (2013.01); ***H04L 63/107*** (2013.01)

(58) **Field of Classification Search**
CPC H04L 63/083; H04L 63/107
See application file for complete search history.

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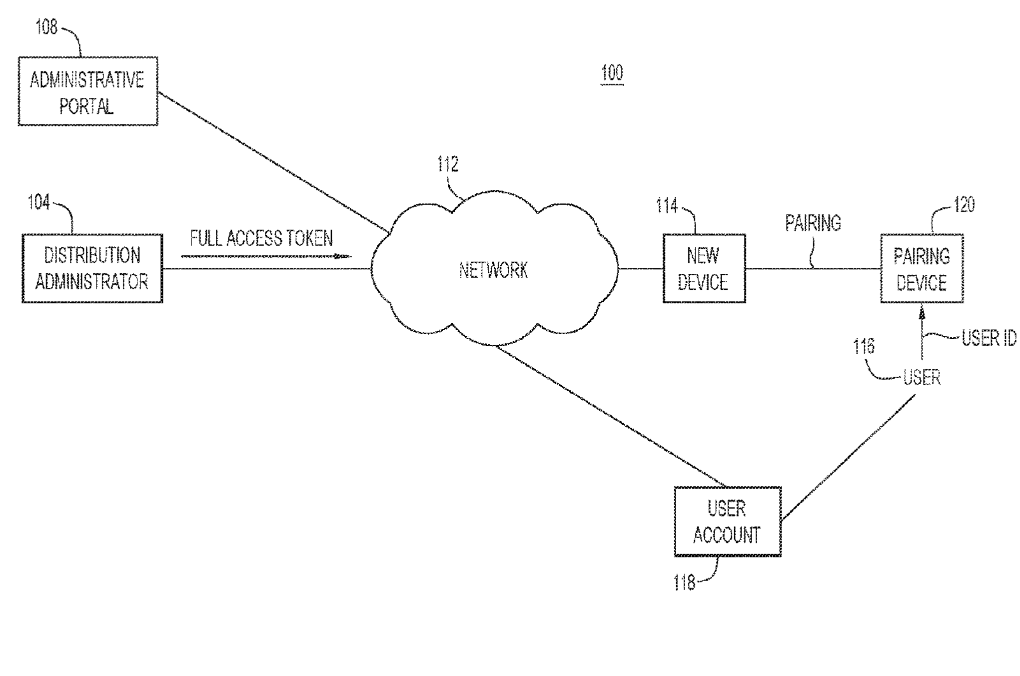
Primary Examiner — Michael R Vaughan
(74) *Attorney, Agent, or Firm* — Edell, Shapiro & Finnan, LLC

(57) **ABSTRACT**

A method comprises: at a control entity configured to communicate with a network: linking a user identifier of a user having an account for call-related information to a hardware identifier of a device to be sent to the user, to produce linked identifiers; loading on the device a limited use token associated with the user that enables only basic call functions on the device, and prevents the device from accessing the account; determining whether the user possesses the device based on the linked identifiers and proximity pairing of the device with a pairing device; and based on determining, either provisioning the device with a full access token that enables full calling functions on the device and grants full access to the account by the device, or not provisioning the device with the full access token to prevent access to the account.

20 Claims, 9 Drawing Sheets



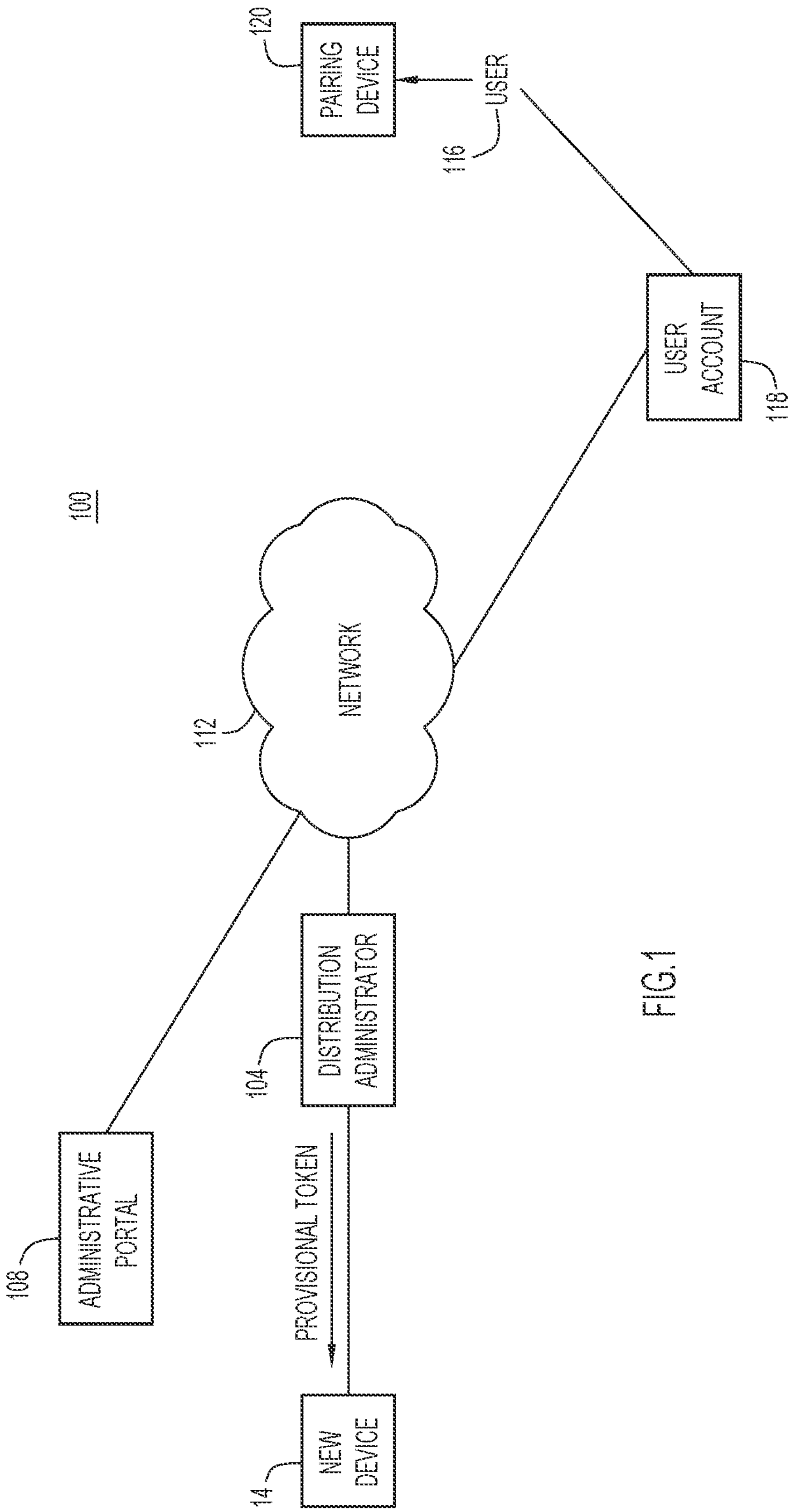

100
PAIRING DEVICE
120
USER
116
USER ACCOUNT
118
NETWORK
112
DISTRIBUTION ADMINISTRATOR
104
ADMINISTRATIVE PORTAL
108
PROVISIONAL TOKEN
NEW DEVICE
114

FIG.1

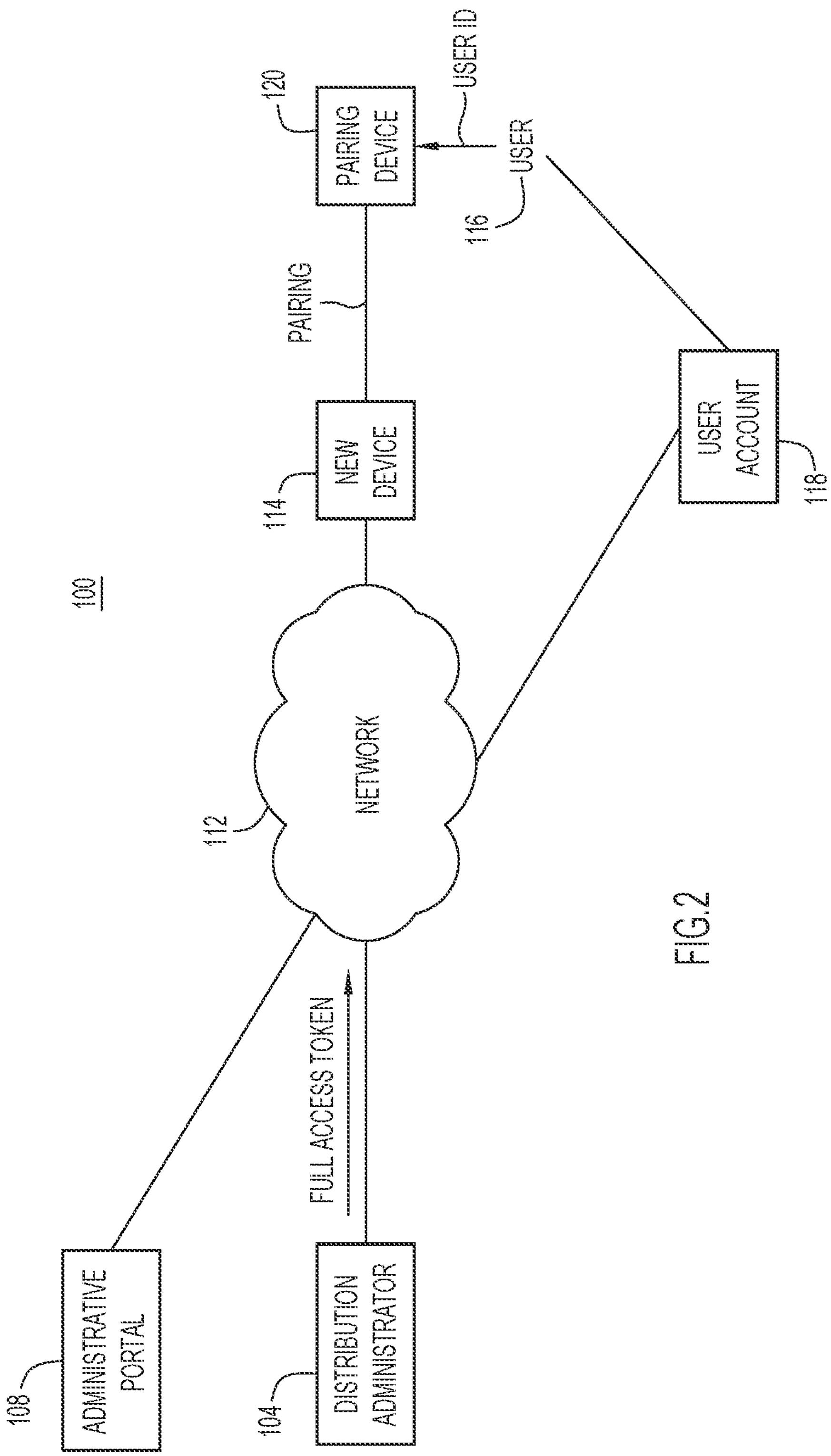

100
120
PAIRING DEVICE
USER ID
USER
116
PAIRING
114
NEW DEVICE
USER ACCOUNT
118
112
NETWORK
FULL ACCESS TOKEN
108
ADMINISTRATIVE PORTAL
104
DISTRIBUTION ADMINISTRATOR

FIG.2

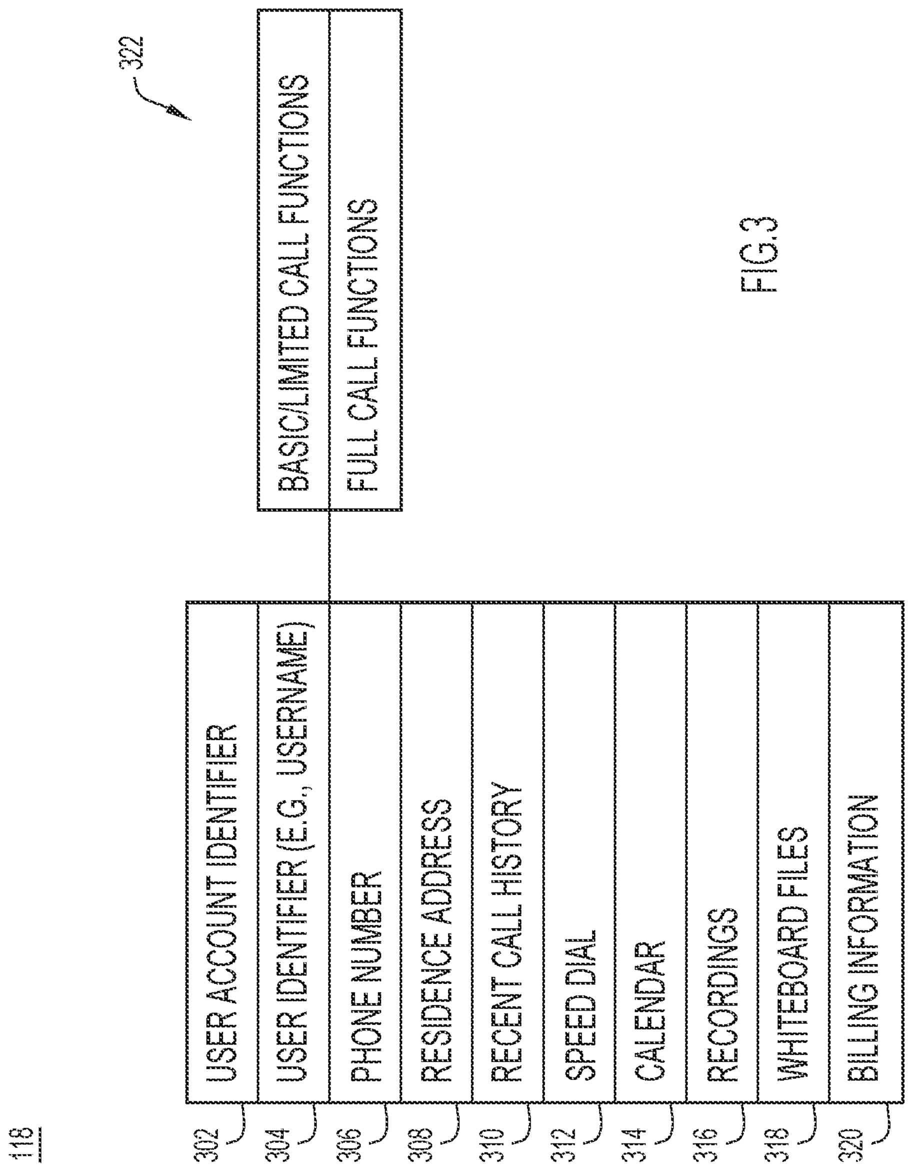

118
322
BASIC/LIMITED CALL FUNCTIONS
FULL CALL FUNCTIONS
302 USER ACCOUNT IDENTIFIER
304 USER IDENTIFIER (E.G., USERNAME)
306 PHONE NUMBER
308 RESIDENCE ADDRESS
310 RECENT CALL HISTORY
312 SPEED DIAL
314 CALENDAR
316 RECORDINGS
318 WHITEBOARD FILES
320 BILLING INFORMATION

FIG.3

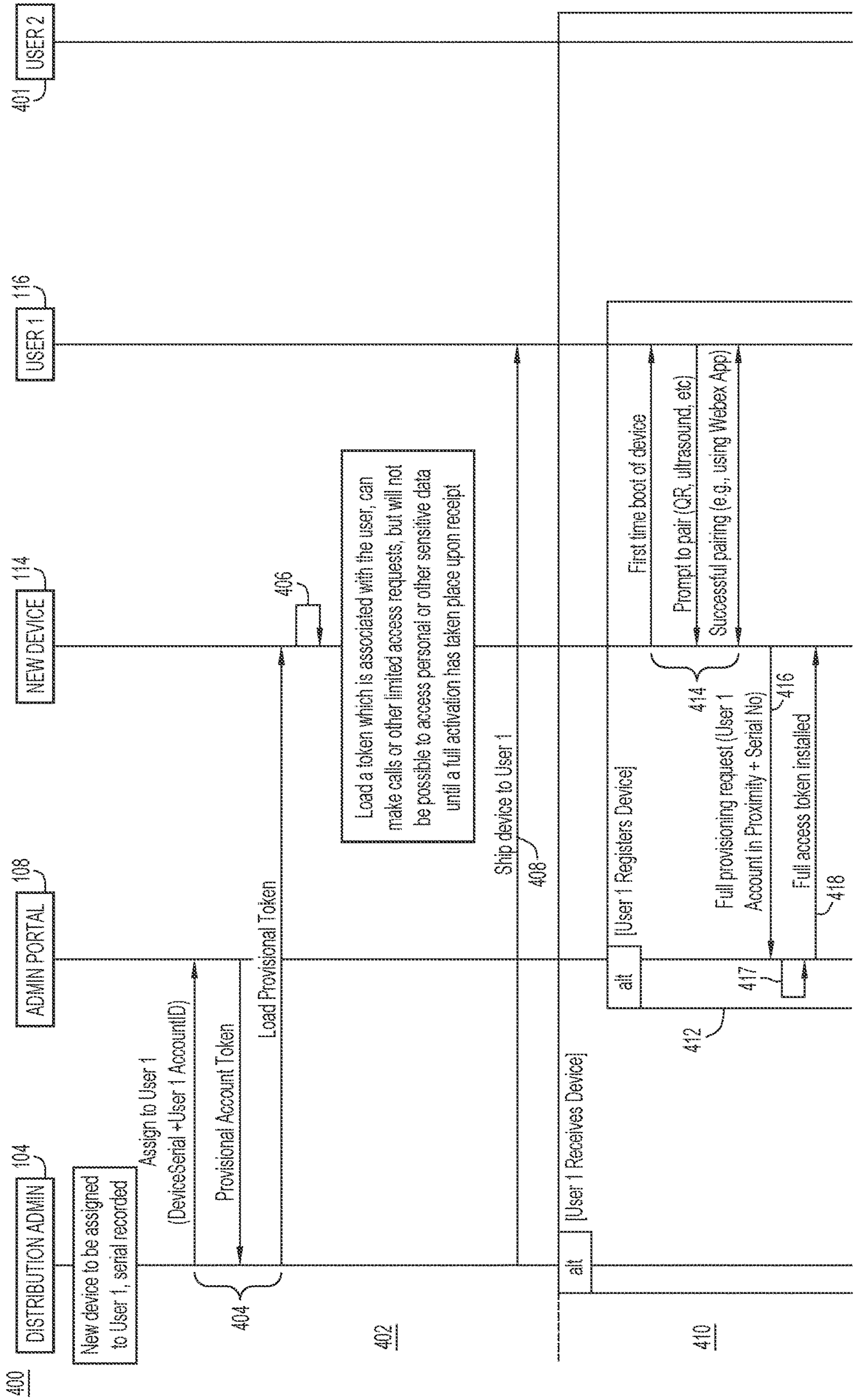

400
DISTRIBUTION ADMIN
104
ADMIN PORTAL
108
NEW DEVICE
114
USER 1
116
USER 2
401
New device to be assigned to User 1, serial recorded
Assign to User 1 (DeviceSerial +User 1 AccountID)
Provisional Account Token
Load Provisional Token
404
402
406
Load a token which is associated with the user, can make calls or other limited access requests, but will not be possible to access personal or other sensitive data until a full activation has taken place upon receipt
Ship device to User 1
408
alt
[User 1 Receives Device]
410
alt
[User 1 Registers Device]
412
First time boot of device
Prompt to pair (QR, ultrasound, etc)
Successful pairing (e.g., using Webex App)
414
Full provisioning request (User 1 Account in Proximity + Serial No)
416
417
Full access token installed
418

FIG.4A

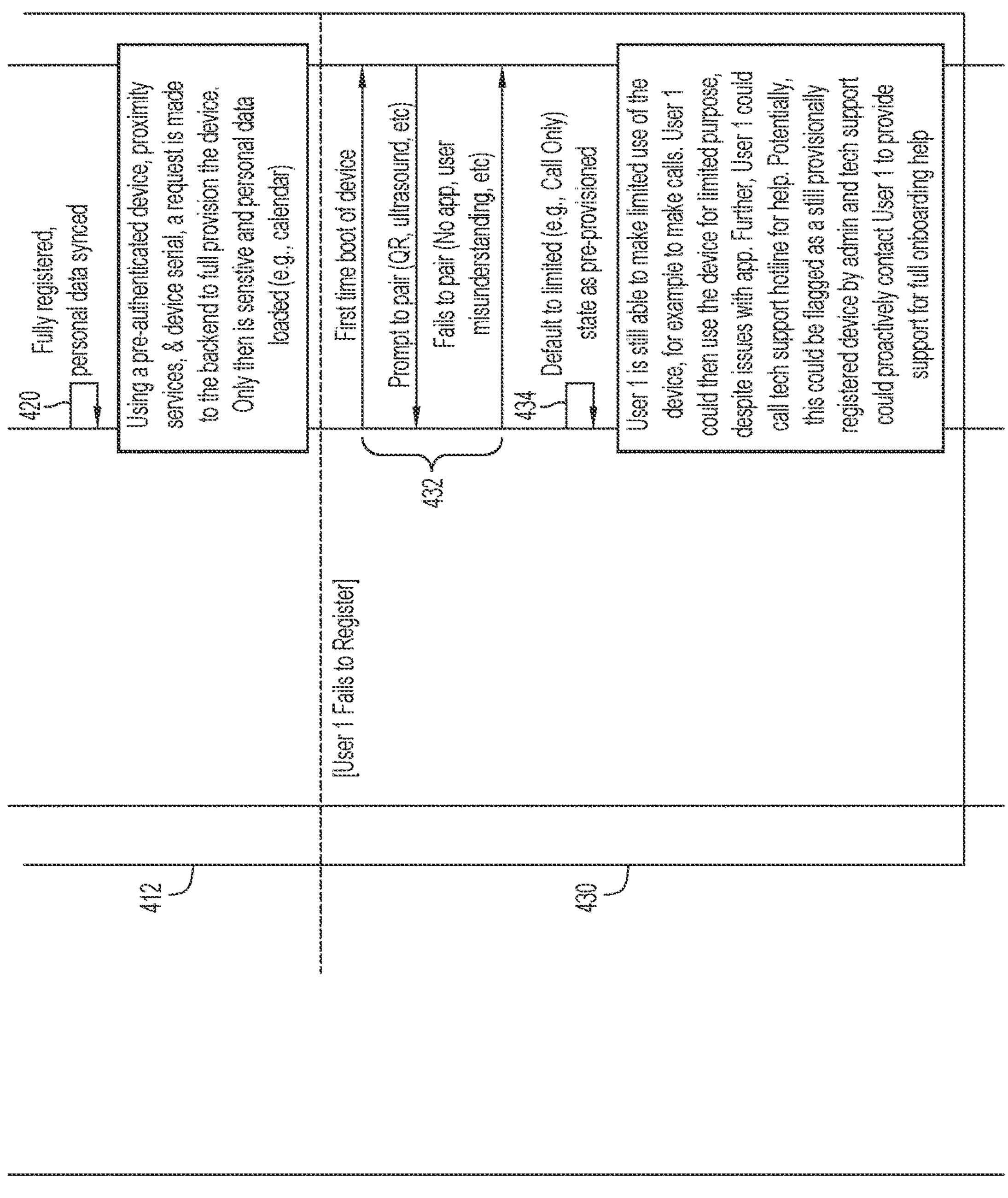

420
Fully registered, personal data synced
Using a pre-authenticated device, proximity services, & device serial, a request is made to the backend to full provision the device. Only then is sensitive and personal data loaded (e.g., calendar)
First time boot of device
Prompt to pair (QR, ultrasound, etc)
Fails to pair (No app, user misunderstanding, etc)
432
434
Default to limited (e.g., Call Only) state as pre-provisioned
User 1 is still able to make limited use of the device, for example to make calls. User 1 could then use the device for limited purpose, despite issues with app. Further, User 1 could call tech support hotline for help. Potentially, this could be flagged as a still provisionally registered device by admin and tech support could proactively contact User 1 to provide support for full onboarding help
[User 1 Fails to Register]
412
430

FIG.4B

410

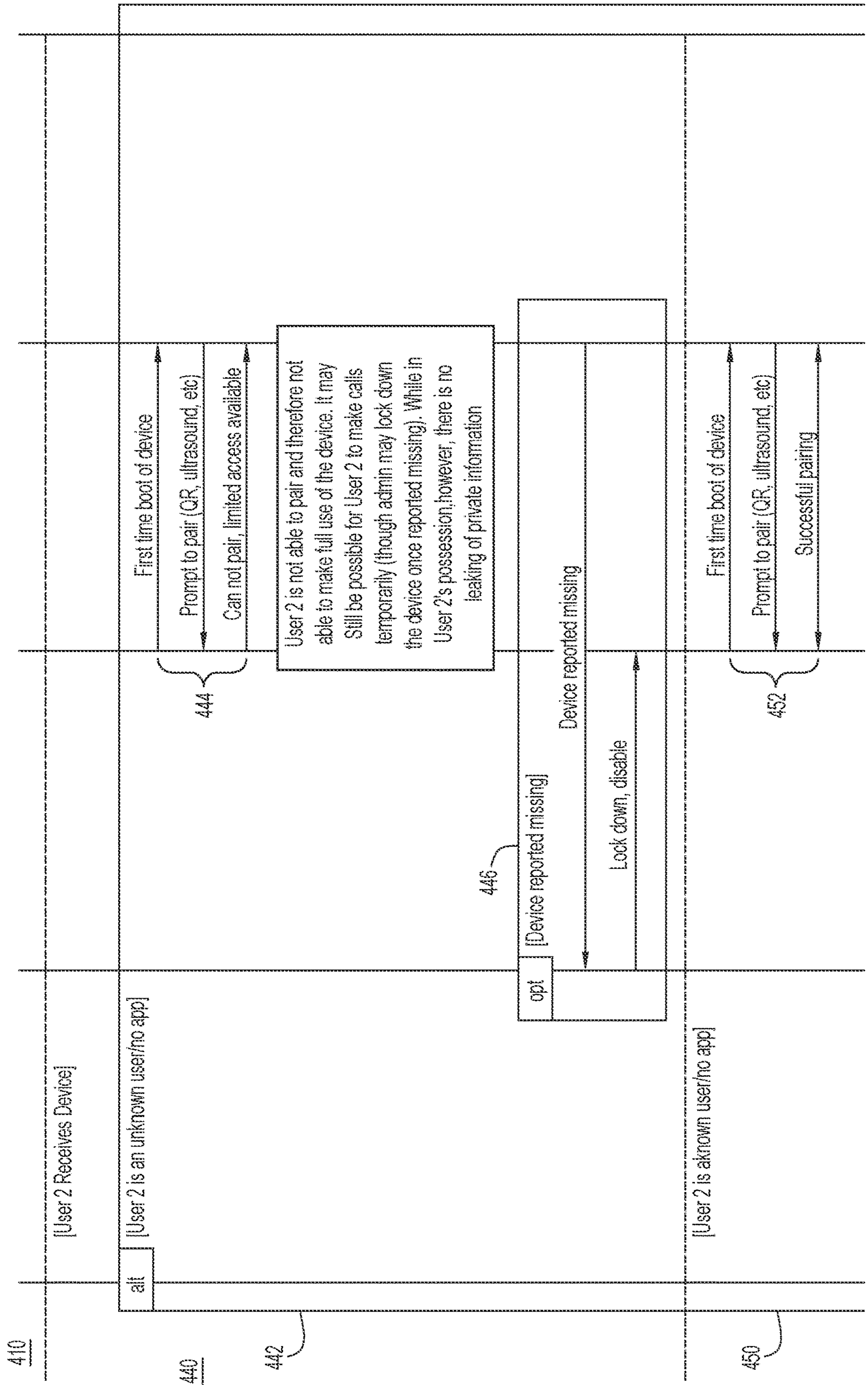

410
440
442
450
alt
[User 2 Receives Device]
[User 2 is an unknown user/no app]
First time boot of device
Prompt to pair (QR, ultrasound, etc)
Can not pair, limited access available
444
User 2 is not able to pair and therefore not able to make full use of the device. It may Still be possible for User 2 to make calls temporarily (though admin may lock down the device once reported missing). While in User 2's possession,however, there is no leaking of private information
opt
[Device reported missing]
446
Device reported missing
Lock down, disable
[User 2 is aknown user/no app]
First time boot of device
Prompt to pair (QR, ultrasound, etc)
Successful pairing
452

FIG.4C

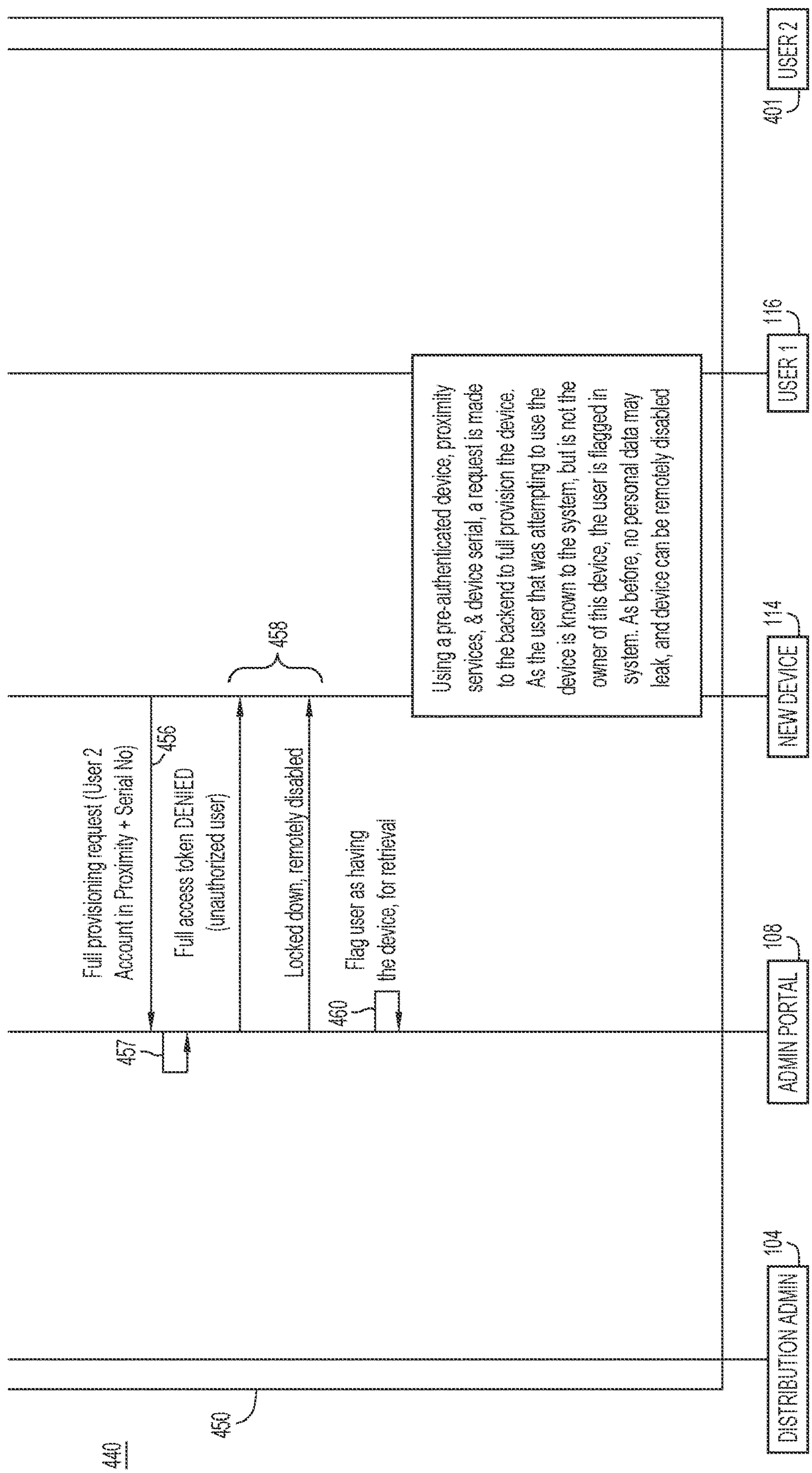

440
450
DISTRIBUTION ADMIN
104
ADMIN PORTAL
108
NEW DEVICE
114
USER 1
116
USER 2
401
Full provisioning request (User 2 Account in Proximity + Serial No)
456
457
Full access token DENIED (unauthorized user)
Locked down, remotely disabled
458
460
Flag user as having the device, for retrieval
Using a pre-authenticated device, proximity services, & device serial, a request is made to the backend to full provision the device. As the user that was attempting to use the device is known to the system, but is not the owner of this device, the user is flagged in system. As before, no personal data may leak, and device can be remotely disabled

FIG.4D

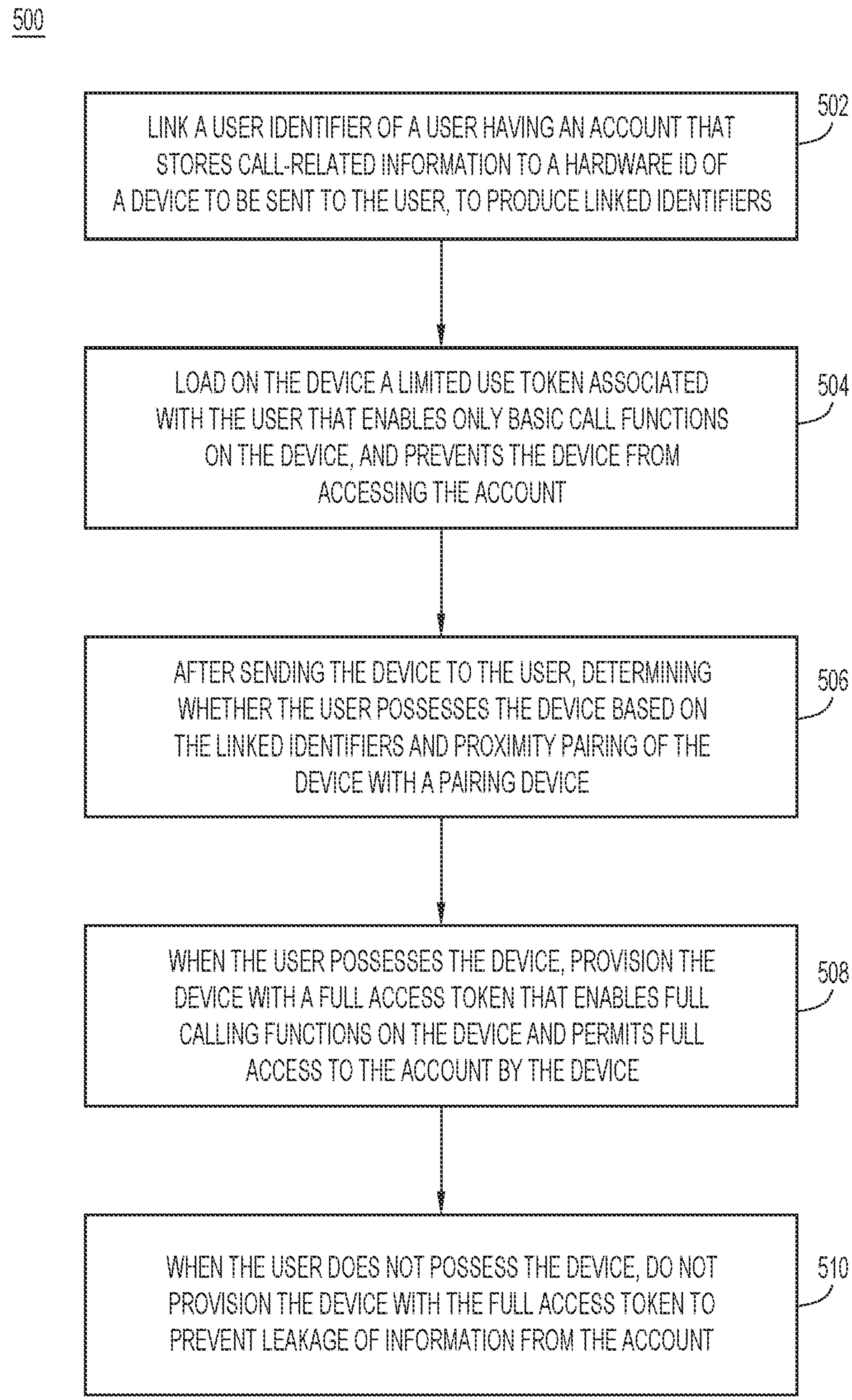

500
LINK A USER IDENTIFIER OF A USER HAVING AN ACCOUNT THAT STORES CALL-RELATED INFORMATION TO A HARDWARE ID OF A DEVICE TO BE SENT TO THE USER, TO PRODUCE LINKED IDENTIFIERS
502
LOAD ON THE DEVICE A LIMITED USE TOKEN ASSOCIATED WITH THE USER THAT ENABLES ONLY BASIC CALL FUNCTIONS ON THE DEVICE, AND PREVENTS THE DEVICE FROM ACCESSING THE ACCOUNT
504
AFTER SENDING THE DEVICE TO THE USER, DETERMINING WHETHER THE USER POSSESSES THE DEVICE BASED ON THE LINKED IDENTIFIERS AND PROXIMITY PAIRING OF THE DEVICE WITH A PAIRING DEVICE
506
WHEN THE USER POSSESSES THE DEVICE, PROVISION THE DEVICE WITH A FULL ACCESS TOKEN THAT ENABLES FULL CALLING FUNCTIONS ON THE DEVICE AND PERMITS FULL ACCESS TO THE ACCOUNT BY THE DEVICE
508
WHEN THE USER DOES NOT POSSESS THE DEVICE, DO NOT PROVISION THE DEVICE WITH THE FULL ACCESS TOKEN TO PREVENT LEAKAGE OF INFORMATION FROM THE ACCOUNT
510

FIG.5

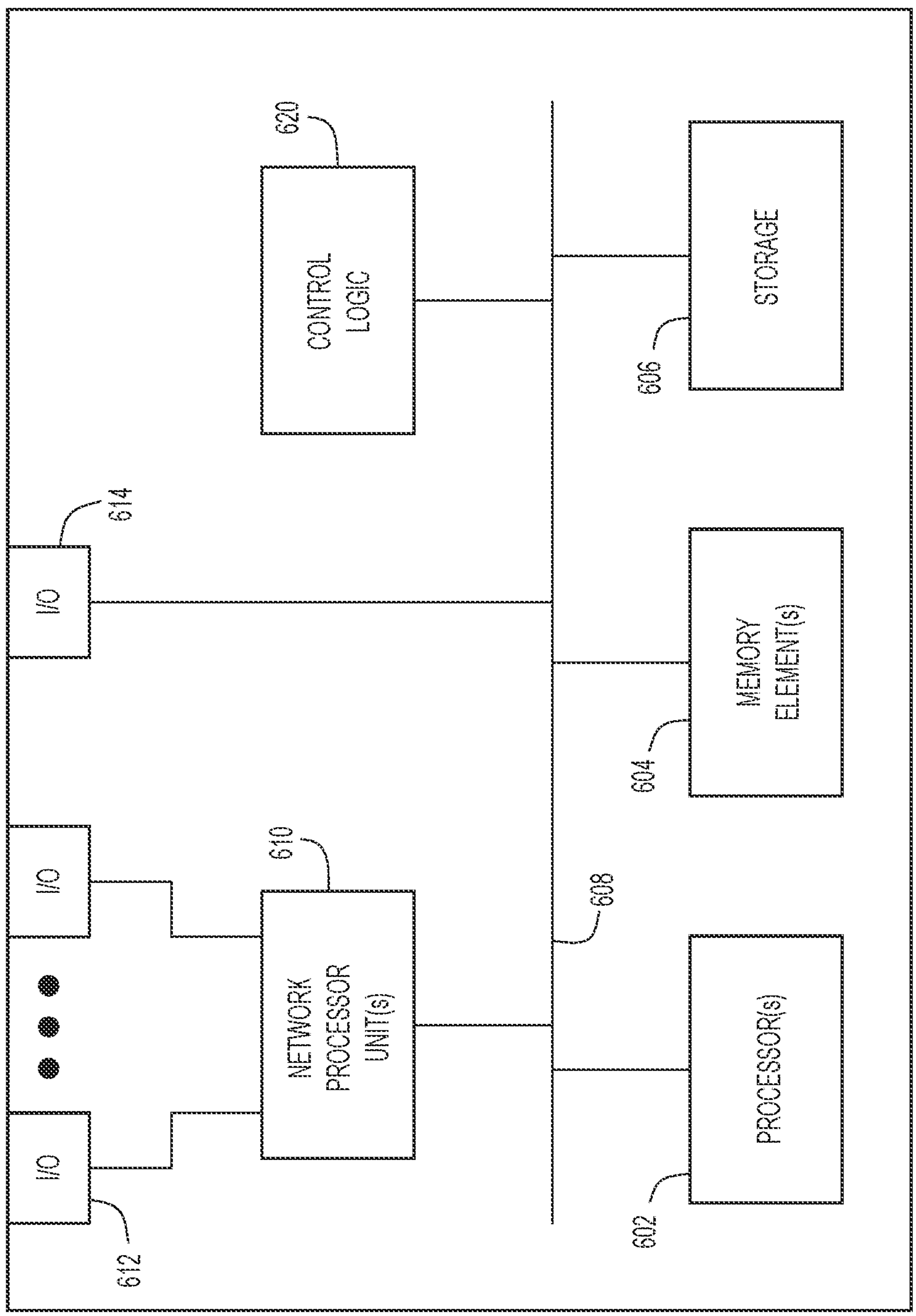

CONTROL LOGIC
620
STORAGE
606
I/O
614
MEMORY ELEMENT(s)
604
I/O
610
NETWORK PROCESSOR UNIT(s)
608
I/O
612
PROCESSOR(s)
602

FIG.6

600

PREVENTING DATA AND HARDWARE LOSS IN PRE-PROVISIONED DEVICES

TECHNICAL FIELD

The present disclosure relates to deploying pre-provisioned new communication devices to intended end users.

BACKGROUND

In the past, communication devices such as phones, conference room systems, and other hardware may be shipped in an unconfigured state, i.e., in a factory reset state without user, line, or other account provisioning. Sometimes, the unconfigured devices are shipped to information technology (IT) departments. There, IT personnel install, setup, and manage full user provisioning of the communication devices, and then ship the fully configured communication devices to end users. In another scenario, the communication devices are shipped directly to hybrid users at their residences, but the users may be unfamiliar with how to setup the communication devices or the associated user accounts. In yet another scenario, when shipping the communication devices to users, service providers may send factory reset phones to a distribution partner that fully pre-provisions network and account data, to avoid having the users and administrators setup the communication devices. The distribution partner then ships the fully pre-provisioned communication devices directly to the users.

While this may be convenient for the users who receive fully configured communication devices, problems may arise when the communication devices are inadvertently sent to wrong addresses, lost during shipment, or stolen/intercepted. The fully configured communication devices can permit access to personal/sensitive account information, such as recent call history, calendar data, whiteboard information, billing information, and the like. Such access to personal data is a concern when unauthorized users gain possession of the communication devices, and can access the information, which represents unauthorized leakage of personal data.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1** is a block diagram of a network environment in which pre-provisioning new devices to prevent personal data leakage during deployment of the new devices to intended users may be implemented, according to an example embodiment.

FIG. **2** shows the network environment as configured after shipment of a new device to an intended user of the new device, according to an example embodiment.

FIG. **3** is an illustration of account information stored in a user account for the intended user of the new device, according to an example embodiment.

FIGS. **4**A, **4**B, **4**C, and **4**D collectively represent a diagram of transactions performed in the network environment to pre-provision, ship, and post-shipment provision the new device for the intended user, according to an example embodiment.

FIG. **5** is a method of pre-provisioning a new device to prevent personal data leakage during deployment of the new device to the intended user, according to an example embodiment.

FIG. **6** illustrates a hardware block diagram of a computing device that may perform functions associated with operations performed in the embodiments presented herein, according to an example embodiment.

DETAILED DESCRIPTION

Overview

In an embodiment, a method comprises: at a control entity configured to communicate with a network: linking a user identifier of a user having an account for call-related information to a hardware identifier of a device to be sent to the user, to produce linked identifiers; loading on the device a limited use token associated with the user that enables only basic call functions on the device, and prevents the device from accessing the account; determining whether the user possesses the device based on the linked identifiers and proximity pairing of the device with a pairing device; and based on determining, either provisioning the device with a full access token that enables full calling functions on the device and grants full access to the account by the device, or not provisioning the device with the full access token to prevent full access to the account.

Example Embodiments

FIG. **1** is a block diagram of an example network environment **100** in which embodiments directed to pre-provisioning new devices to prevent personal data leakage during deployment of the new devices to intended users may be implemented. The embodiments ensure that the intended users receive the new devices with proper user account provisioning and without leaking personal data from user accounts of the intended end users. The embodiments also reduce the possibility that the new devices become lost after shipment. The embodiments achieve the foregoing in a way that limits setup complexity for the intended users, so that the intended users can setup the new devices quickly "out of the box," and then begin using the new devices.

Network environment **100** includes a distribution administrator **104** and an administrative portal **108** each connected to a network **112** and configured to communicate with each over the network. Network **112** may include wide area networks (WANs), such as the Internet, and local area networks (LANs), cellular/mobile networks, Wi-Fi networks, and so on. Network **112** forwards traffic between endpoints in the form of data packets, e.g., Internet Protocol (IP) packets, in accordance with known or hereafter developed protocols, such as the Transmission Control Protocol (TCP)/IP (TCP/IP), and the like.

Distribution administrator **104** serves as an administrator for a device distribution center that stores or warehouses new devices intended for geographically distributed users. Each new device may be a wired and/or wireless network communication end user device, such as a cell phone/smartphone, an IP phone, a video conference device or endpoint, and the like. When initially stored with distribution administrator **104**, the new devices are generally not provisioned or configured for users, but are uniquely identified by hardware identifiers, such as hardware serial numbers, programmed into memory of the user devices and accessible through the user devices, as is known. In the example of FIG. **1**, the device distribution center warehouses a new device **114** intended for a user **116** (i.e., the "intended user," also referred to as "User1").

User **116** has a user account **118** (i.e., a cloud-based user account) stored in one or more cloud repositories/datacenters accessible through network **112**, such as in a datastore associated with distribution administrator **104**, administrative portal **108**, service providers that submit orders to the distribution administrator to ship new devices to users, corporate entities/enterprise, and the like. User **116** can access user account **118** through new device **114** once the user possesses the new device and administrative portal **108** has enabled the new device for account access according to embodiments presented herein. User account **118** may define call functions on new device **114** and include account information specific to user **116**. The account information may be sensitive/private information of user **116**, and therefore should only be accessible to the user (i.e., the intended or authorized user), and no other user.

User **116** has access to/operates a pairing device **120** that is local to the user. Pairing device **120** may include a computer device, such as a cell phone/smartphone, a computer (e.g., laptop, tablet, and so on), and may also host conferencing applications (such as Webex), quick response (QR) applications, and ultrasound applications that offer proximity paring capabilities, for example. When operated by user **116**, pairing device **120** obtains/acquires information from the user that identifies the user (referred to as a "user identifier"). For example, pairing device **120** may acquire the user identifier (ID) (userID) when the user performs a one-time login to/on the pairing device during which the user provides the userID to the pairing device.

According to embodiments presented herein, new device **114** intended for user **116** is pre-provisioned, shipped to the user, and then enabled for all call functions and full access to user account **118** (access to full/all information of the account), under certain conditions, described below. At a high-level, the embodiments include a pre-delivery phase that occurs before new device **114** is shipped to user **116**, shipment of the user device to the user, and a post-delivery phase that occurs after the new device has been shipped to the user. During the pre-delivery phase, distribution administrator **104** and administrative portal **108** (collectively referred to as a "control entity" or a "device deployment manager") interact to pre-provision new device **114** with a provisional token (also referred to a "limited use token") that is linked to/associated with user **116**. The limited use token enables basic call functions only on new device **114**, and prevents the new device from accessing user account **118**.

After new device **114** has been shipped to user **116**, the post-delivery phase ensues. FIG. **2** shows network environment **100** after shipment of new device **114** to user **116**. In the post-delivery phase, new device **114** pairs with pairing device **120** as shown in FIG. **2**, and acquires information identifying user **116** from the pairing device. Next, new device **114** reports to administrative portal **108** information identifying new device **114** and the information identifying user **116** acquired through pairing with pairing device **120**. Administrative portal **108** confirms/verifies that user **116** actually possesses new device **114** based on the reported information, and also verifies that the user is authorized to possess the new device.

Upon confirming the foregoing, administrative portal **108** provisions new device **114** with a full access token (also referred to as a "fully scoped token") that overrides the limited use token. The full access token enables full call functions on new device **114**, and grants to the new device full access to user account **118**. On the other hand, when the confirmation fails, and administrative portal **108** learns that user **116** does not possess new device **114**, administrative portal **108** does not provision the new device with the full access token, which prevents the new device from accessing user account **118** and avoids leakage of account information therein to the new device and any unauthorized user that may be in possession of the new device. Additionally, administrative portal **108** may remotely disable new device **114**, and send an alarm to an administrator.

FIG. **3** is an illustration of user account **118**, according to an embodiment. User account **118** includes a user account identifier **302**, a userID **304** (e.g., a username and/or user email address, and so on), a phone number **306**, a residence address **308**, recent call history **310** (e.g., a list of recently called/received phone numbers), speed dial **312**, calendar **314** (e.g., scheduled meetings and participants), recordings **316** (e.g., conference call recordings), whiteboard files **318**, and billing information **320** (e.g., billing account information, usage, and so on). In addition, user account **118** may include call definitions **322** that define basic call functions (e.g., incoming and outgoing calling only for certain phone numbers), and full call functions (e.g., unlimited incoming and outgoing calling for unlimited phone numbers).

FIGS. **4A-4D** are a diagram of example transactions **400** performed in network environment **100** to pre-provision, ship, and post-shipment provision (i.e., fully configure/enable) new device **114** for user **116** (User1—the intended user). Transactions **400** (which also include operations) include interactions between distribution administrator **104** ("DistributionAdmin"), administrative portal **108** ("Admin-Portal"), new device **114**, user **116** (i.e., intended/authorized User1), and a user **401** (also referred to as "User2"). User2 is an unintended/unauthorized user. In FIGS. **4A-4D**, "User1" and "User2" may refer to the individuals User1 and User2 and/or pairing devices operated by the individuals, depending on context.

With reference to FIG. **4A**, transactions **400** begin with the following pre-shipment/pre-provisioning transactions **402**. At **404**, upon receiving (e.g., from a service provider) an order to assign new device **114** to User1, distribution administrator **104** provides to administrative portal **108** a hardware ID (e.g., hardware serial number) of new device **114** and a userID of User1. The userID may be any of a username, email address, and a user account ID of User1. In response, distribution administrator **104** links/associates the userID to the hardware ID to form a linked tuple of the identifiers. For example, distribution administrator **104** stores the userID and the hardware ID together in memory with address pointers to each other. The foregoing operation links/associates new device **114** to User1 (i.e., to the identity of User1) at administrative portal **108**.

Administrative portal **108** provides to distribution administrator **104** a provisional/limited use token associated with/linked to User1, as described above. In response, distribution administrator **104** installs/loads the provisional token on new device **114**. For example, administrative portal may program new device **114** with the provisional token through a communication/configuration port of the new device. Responsive to installation of the provisional token, at **406**, new device **114** becomes provisionally activated, meaning that the new device can perform basic call functions, and may provide limited status to administrative portal **108**, but cannot access user account **118**. The basic call functions may be limited to receiving phone calls, calling an emergency service number, and calling a preloaded IT help line for user support for activating the device, for example. Distribution administrator **104** may also install on new device **114** a pairing application that may be invoked to pair the new device with pairing device **120**, as described below.

At **408**, distribution administrator **104** ships new device **114** to User1. Post-shipment provisioning transactions are described below.

The post-shipment provisioning transactions include first alternative transactions **410** that occur when User1 (the intended user) receives new device **114**. First alternative transactions **410** include the following first scenario transactions **412** that occur when User1 successfully registers new device **114**. At **414**, new device **114** successfully pairs with pairing device **120**, which is near the new device. To do this, User1 powers-on/boots-up new device **114**, which presents a prompt to pair the new device. Responsive to the prompt, User1 initiates pairing of new device**114** with pairing device **120**, which leads to successful pairing of the two devices. Any known or hereafter developed pairing technique may be employed. In one example, the two devices perform ultrasound pairing, i.e., exchange ultrasound signals to pair with each other. In another example, the two devices may exchange a preconfigured QR code to pair with each other. New device **114** acquires information identifying User1 (i.e., the userID for User1) through pairing device **120**. The userID may be in the form of the username, the email address, or the user account ID of User1. After pairing, at **416**, new device **114** reads its hardware ID and sends to administrative portal **108** a full provisioning request that includes/reports the hardware ID of new device **114** (also referred to as the "reported hardware ID") and the userID for User1 (also referred to as the "reported userID") acquired through pairing.

Upon receiving the full provisioning request from new device **114**, at **417**, administrative portal **108** determines whether the reported hardware ID and the reported userID respectively match the previously linked hardware ID and userID. A match between the reported userID and the linked userID is confirmed when the reported userID is any reported form of userID for User1 (e.g., email, username, or user account ID) and the linked userID is also any form of userID for User1 (e.g., email, username, or user account ID). For example, a match between userIDs is confirmed even when the reported userID for User1 is a username and the linked userID for User1 is an account ID for User1.

When there is a match, at **418**, administrative portal **108** configures new device **114** with a full access token linked to/associated with User1. That is, administrative portal **108** sends the full access token to new device **114**, and the new device installs the full access token. The full access token overrides the limited use token. The full access token enables full call functions on new device **114**. The full access token also grants to new device **114** full access to user account **118**, i.e., to all the account information stored in the user account. Referring to FIG. **4**B, using the full access token, at **420**, new device **114** synchronizes to (e.g., downloads at least some of) the account information in user account **118**. In an example, the full access token may be configured as an Oauth token that can be used to access account information from user account **118** using Oauth request/response techniques.

When there is not a match between the reported hardware ID and userID and the linked hardware ID and userID, respectively, administrative portal **108** does not send the full access token to new device **114**, which prevents the new device from accessing user account **118** and leaking sensitive personal data of User1.

First alternative transactions **410** include the following second scenario transactions **430** that occur when User1 fails to register new device **114**. At **432**, new device **114** fails to pair with any pairing device (e.g., with pairing device **120**). For example, User1 powers-on/boots-up new device **114**, and the new device presents the prompt to pair the new device. In this case, new device **114** fails to pair with any pairing device, and as a result the new device does not receive the full access token from administrative portal **108**. At **434**, new device **114** remains limited to the basic call functions, and is unable to fully access user account **118**. Using the basic call functions on new device **114**, User1 may call IT support for guidance on registering the new device. Alternatively, administrative portal **108** may flag new device **114** as being provisionally registered, and submit a request to IT support to call User1 on the new device.

With reference to FIG. **4**C, the post-shipment provisioning transactions also include second alternative transactions **440** that occur when unauthorized User2 receives new device **114** instead of User1. Second alternative transactions **440** include the following third scenario transactions **442** that occur when User2 is an unknown user (i.e., is unknown to administrative portal **108**). At **444**, new device **114** fails to pair with any pairing device. For example, User2 powers-on/boots-up new device **114**, and the new device presents the pairing prompt mentioned above. In this case, new device **114** fails to pair with any pairing device, and the new device does not receive the full access token from administrative portal **108**. New device **114** remains limited to the basic calling functions, and is unable to access user account **118**. Thus, User2 may make use new device **114** for the basic call functions, but cannot access user account **118**.

In another option, at **446**, User1 reports to administrative portal **108** that User1 did not receive new device **114**, i.e., that the new device is missing. In response, administrative portal **108** sends a lock-down command to new device **114** that disables (i.e., locks-down) the new device. In other words, administrative portal **108** remotely disables new device **114**. Upon receiving the lock-down command, new device **114** locks itself.

Second alternative transactions **440** include the following fourth scenario transactions **450** that occur when User2 is a known user (i.e., administrative portal recognizes User2). At **452**, new device **114** successfully pairs with a pairing device operated by User2 that is different from pairing device **120**. For example, User2 powers-on/boots-up new device **114**, and the new device presents the pairing prompt. Responsive to the prompt, User2 initiates pairing of new device **114** with the nearby pairing device operated by User2, which leads to successful pairing of the two devices. New device **114** acquires a userID for User2 (which is different from the userID for User1) from the pairing device. With reference to FIG. **4**D, after pairing, at **456**, new device **114** sends to administrative portal **108** a full provisioning request that includes/reports the hardware ID of new device **114** and the userID for User2 acquired through pairing.

Upon receiving the full provisioning request from new device **114**, at **457**, administrative portal **108** determines that the reported hardware ID for new device **114** and the reported userID for User2 do not match the previously linked hardware ID for the new device and the userID for User1. In this case, there is no match because the userIDs differ-User2 is an unauthorized user for new device **114**. Responsive to the failure to match, at **458**, administrative portal **108** denies the full access token and disables/locks-down new device **114**. At **460**, administrative portal **108** flags User2 (who is known) as having new device **114**, and submits an order to retrieve the new device from User2.

In the scenarios described above, a mismatch between the reported userID and the userID initially linked to the hardware identifier of the new device represents an invalid activation attempt. Upon detecting the invalid activation attempt, administrative portal **108** provides an alarm to an administrator and/or the user (i.e., the intended user) indicating that an attempt has been made to activate the new device by an unauthorized user. The alarm may indicate an IP address of the new device, the hardware ID of the new device, the username of the user, an organization ID of the user, and an organization ID or other userID of the unauthorized user, if known. Additionally, a device management rule configured on administrative portal **108** may be triggered to fully lock the new device.

FIG. **5** is a flowchart of an example method **500** of pre-provisioning a new device (also referred to simply as a "device") to prevent personal account data leakage during deployment of the device to a user intended (i.e., an intended end user) for the device. Method **500** may be performed in network environment **100**, primarily by administrative portal **108** operating in connection with distribution administrator **104** (collectively referred to as a "control entity" or "back-end") and the new device. In the ensuing description of FIG. **5**, the new device is referred to simply as the "device."

During a pre-provisioning phase that occurs before the device is sent to the user, at **502**, the control entity links/associates information identifying the user to a hardware identifier of the device to produce linked identifiers. The user has a cloud-based account (i.e., a user account) that stores personal and call-related account information. In an example, the control entity links a user identifier of the user (e.g., a username, an account identifier, or an email address) to the hardware identifier.

Also during the pre-provisioning phase, at **504**, the control entity load/installs/programs into memory of the device a limited use token associated with (i.e., linked to) the user that enables only basic call functions on the device, and prevents the device from accessing full (or, in some cases, any) information from the account. The control entity may also install on the device a proximity pairing application that enables proximity pairing of the device with a pairing device near the device, based on an exchange between the two devices of one of a QR code, pairing ultrasound signals, and the like.

After the device has been sent to the user (e.g., after sending the device to the user), at **506**, the control entity determines whether the user possesses the device by comparing the linked identifiers against information obtained from the device (e.g., reported identifiers) from proximity pairing the device with the pairing device that is nearby. The pairing device has access to information indicative of a user identifier of a user, which may or may not be the intended user.

To determine whether the user possesses the device, the control entity may perform the following operations. The control entity receives from the device a full provisioning request (sent by the device) that carries a reported hardware identifier and a reported user identifier acquired by the device from the proximity pairing. In response, the control entity determines/evaluates whether the reported hardware identifier and the reported user identifier match the linked hardware identifier and user identifier. When there is a match, the control entity confirms that the user possesses the device. When there is not a match, the control entity confirms that the user does not possess the device.

When the user possesses the device, at **508**, the control entity provisions the device with a full access token that enables full call functions on the device and grants/permits full access to the account (i.e., full account access) by the device. The full call functions include receiving and making calls from and to any number.

On the other hand, when the user does not possess the device, at **510**, the control entity does not provision the device with the full access token in order to prevent leakage of information from the account. The control function may remotely disable the device. The control function may provide an alarm to an administrator indicating that an attempt has been made to activate the device by an unauthorized user. The alarm may also indicate an identity of the user, an IP address of the device, and so on.

In summary, according to embodiments presented herein, a control entity pre-associates a hardware identifier of a new device (that is to be shipped to a user) with a user account of the user. Before shipping the new device to the user, the control entity installs on the new device only a provisional/limited use token associated with the user. The limited use token prevents the new device from accessing the user account. After the control entity ships the new device to the user, the control entity exchanges information with the new device, and access the pre-associated information, to verify that the user has paired the new device with a nearby device that has access to information identifying the user. When verifying is successful, the control entity grants all user account access to the new device. When verifying is unsuccessful, the control entity does not grant user account access to the new device, which prevents user account data leakage. The control entity alerts administrators that the new device may be intercepted or otherwise subject to unauthorized use. The embodiments improve upon existing pre-provisioning models for new devices.

Referring to FIG. **6**, FIG. **6** illustrates a hardware block diagram of a computing device **600** that may perform functions associated with operations discussed herein in connection with the techniques depicted in FIGS. **1-5**. In various embodiments, a computing device or apparatus, such as computing device **600** or any combination of computing devices **600**, may be configured as any entity/entities as discussed for the techniques depicted in connection with FIGS. **1-5** in order to perform operations of the various techniques discussed herein. For example, computing device may represent each of the entities of network environment **100**, including administrative portal **108**, distribution administrator **104**, the control entity/device deployment entity, new device **114**, and pairing device **120** individually and/or collectively.

In at least one embodiment, the computing device **600** may be any apparatus that may include one or more processor(s) **602**, one or more memory element(s) **604**, storage **606**, a bus **608**, one or more network processor unit(s) **610** interconnected with one or more network input/output (I/O) interface(s) **612**, one or more I/O interface(s) **614**, and control logic **620**. In various embodiments, instructions associated with logic for computing device **600** can overlap in any manner and are not limited to the specific allocation of instructions and/or operations described herein.

In at least one embodiment, processor(s) **602** is/are at least one hardware processor configured to execute various tasks, operations and/or functions for computing device **600** as described herein according to software and/or instructions configured for computing device **600**. Processor(s) **602** (e.g., a hardware processor) can execute any type of instructions associated with data to achieve the operations detailed herein. In one example, processor(s) **602** can transform an element or an article (e.g., data, information) from one state or thing to another state or thing. Any of potential processing elements, microprocessors, digital signal processor, baseband signal processor, modem, PHY, controllers, systems, managers, logic, and/or machines described herein can be construed as being encompassed within the broad term 'processor'.

In at least one embodiment, memory element(s) **604** and/or storage **606** is/are configured to store data, information, software, and/or instructions associated with computing device **600**, and/or logic configured for memory element(s) **604** and/or storage **606**. For example, any logic described herein (e.g., control logic **620**) can, in various embodiments, be stored for computing device **600** using any combination of memory element(s) **604** and/or storage **606**. Note that in some embodiments, storage **606** can be consolidated with memory element(s) **604** (or vice versa), or can overlap/exist in any other suitable manner.

In at least one embodiment, bus **608** can be configured as an interface that enables one or more elements of computing device **600** to communicate in order to exchange information and/or data. Bus **608** can be implemented with any architecture designed for passing control, data and/or information between processors, memory elements/storage, peripheral devices, and/or any other hardware and/or software components that may be configured for computing device **600**. In at least one embodiment, bus **608** may be implemented as a fast kernel-hosted interconnect, potentially using shared memory between processes (e.g., logic), which can enable efficient communication paths between the processes.

In various embodiments, network processor unit(s) **610** may enable communication between computing device **600** and other systems, entities, etc., via network I/O interface(s) **612** (wired and/or wireless) to facilitate operations discussed for various embodiments described herein. In various embodiments, network processor unit(s) **610** can be configured as a combination of hardware and/or software, such as one or more Ethernet driver(s) and/or controller(s) or interface cards, Fibre Channel (e.g., optical) driver(s) and/or controller(s), wireless receivers/transmitters/transceivers, baseband processor(s)/modem(s), and/or other similar network interface driver(s) and/or controller(s) now known or hereafter developed to enable communications between computing device **600** and other systems, entities, etc. to facilitate operations for various embodiments described herein. In various embodiments, network I/O interface(s) **612** can be configured as one or more Ethernet port(s), Fibre Channel ports, any other I/O port(s), and/or antenna(s)/antenna array(s) now known or hereafter developed. Thus, the network processor unit(s) **610** and/or network I/O interface(s) **612** may include suitable interfaces for receiving, transmitting, and/or otherwise communicating data and/or information in a network environment.

I/O interface(s) **614** allow for input and output of data and/or information with other entities that may be connected to computing device **600**. For example, I/O interface(s) **614** may provide a connection to external devices such as a keyboard, keypad, a touch screen, and/or any other suitable input and/or output device now known or hereafter developed. In some instances, external devices can also include portable computer readable (non-transitory) storage media such as database systems, thumb drives, portable optical or magnetic disks, and memory cards. In still some instances, external devices can be a mechanism to display data to a user, such as, for example, a computer monitor, a display screen, or the like.

In various embodiments, control logic **620** can include instructions that, when executed, cause processor(s) **602** to perform operations, which can include, but not be limited to, providing overall control operations of computing device; interacting with other entities, systems, etc. described herein; maintaining and/or interacting with stored data, information, parameters, etc. (e.g., memory element(s), storage, data structures, databases, tables, etc.); combinations thereof; and/or the like to facilitate various operations for embodiments described herein.

The programs described herein (e.g., control logic **620**) may be identified based upon application(s) for which they are implemented in a specific embodiment. However, it should be appreciated that any particular program nomenclature herein is used merely for convenience; thus, embodiments herein should not be limited to use(s) solely described in any specific application(s) identified and/or implied by such nomenclature.

In various embodiments, any entity or apparatus as described herein may store data/information in any suitable volatile and/or non-volatile memory item (e.g., magnetic hard disk drive, solid state hard drive, semiconductor storage device, random access memory (RAM), read only memory (ROM), erasable programmable read only memory (EPROM), application specific integrated circuit (ASIC), etc.), software, logic (fixed logic, hardware logic, programmable logic, analog logic, digital logic), hardware, and/or in any other suitable component, device, element, and/or object as may be appropriate. Any of the memory items discussed herein should be construed as being encompassed within the broad term 'memory element'. Data/information being tracked and/or sent to one or more entities as discussed herein could be provided in any database, table, register, list, cache, storage, and/or storage structure: all of which can be referenced at any suitable timeframe. Any such storage options may also be included within the broad term 'memory element' as used herein.

Note that in certain example implementations, operations as set forth herein may be implemented by logic encoded in one or more tangible media that is capable of storing instructions and/or digital information and may be inclusive of non-transitory tangible media and/or non-transitory computer readable storage media (e.g., embedded logic provided in: an ASIC, digital signal processing (DSP) instructions, software [potentially inclusive of object code and source code], etc.) for execution by one or more processor(s), and/or other similar machine, etc. Generally, memory element(s) **604** and/or storage **606** can store data, software, code, instructions (e.g., processor instructions), logic, parameters, combinations thereof, and/or the like used for operations described herein. This includes memory element(s) **604** and/or storage **606** being able to store data, software, code, instructions (e.g., processor instructions), logic, parameters, combinations thereof, or the like that are executed to carry out operations in accordance with teachings of the present disclosure.

In some instances, software of the present embodiments may be available via a non-transitory computer useable medium (e.g., magnetic or optical mediums, magneto-optic mediums, CD-ROM, DVD, memory devices, etc.) of a stationary or portable program product apparatus, downloadable file(s), file wrapper(s), object(s), package(s), container(s), and/or the like. In some instances, non-transitory computer readable storage media may also be removable. For example, a removable hard drive may be used for memory/storage in some implementations. Other examples may include optical and magnetic disks, thumb drives, and smart cards that can be inserted and/or otherwise connected to a computing device for transfer onto another computer readable storage medium.

Variations and Implementations

Embodiments described herein may include one or more networks, which can represent a series of points and/or network elements of interconnected communication paths for receiving and/or transmitting messages (e.g., packets of information) that propagate through the one or more networks. These network elements offer communicative interfaces that facilitate communications between the network elements. A network can include any number of hardware and/or software elements coupled to (and in communication with) each other through a communication medium. Such networks can include, but are not limited to, any local area network (LAN), virtual LAN (VLAN), wide area network (WAN) (e.g., the Internet), software defined WAN (SD-WAN), wireless local area (WLA) access network, wireless wide area (WWA) access network, metropolitan area network (MAN), Intranet, Extranet, virtual private network (VPN), Low Power Network (LPN), Low Power Wide Area Network (LPWAN), Machine to Machine (M2M) network, Internet of Things (IoT) network, Ethernet network/switching system, any other appropriate architecture and/or system that facilitates communications in a network environment, and/or any suitable combination thereof.

Networks through which communications propagate can use any suitable technologies for communications including wireless communications (e.g., 4G/5G/nG, IEEE 802.11 (e.g., Wi-Fi®/Wi-Fi6®), IEEE 802.16 (e.g., Worldwide Interoperability for Microwave Access (WiMAX)), Radio-Frequency Identification (RFID), Near Field Communication (NFC), Bluetooth™, mm.wave, Ultra-Wideband (UWB), etc.), and/or wired communications (e.g., T1 lines, T3 lines, digital subscriber lines (DSL), Ethernet, Fibre Channel, etc.). Generally, any suitable means of communications may be used such as electric, sound, light, infrared, and/or radio to facilitate communications through one or more networks in accordance with embodiments herein. Communications, interactions, operations, etc. as discussed for various embodiments described herein may be performed among entities that may directly or indirectly connected utilizing any algorithms, communication protocols, interfaces, etc. (proprietary and/or non-proprietary) that allow for the exchange of data and/or information.

In various example implementations, any entity or apparatus for various embodiments described herein can encompass network elements (which can include virtualized network elements, functions, etc.) such as, for example, network appliances, forwarders, routers, servers, switches, gateways, bridges, loadbalancers, firewalls, processors, modules, radio receivers/transmitters, or any other suitable device, component, element, or object operable to exchange information that facilitates or otherwise helps to facilitate various operations in a network environment as described for various embodiments herein. Note that with the examples provided herein, interaction may be described in terms of one, two, three, or four entities. However, this has been done for purposes of clarity, simplicity and example only. The examples provided should not limit the scope or inhibit the broad teachings of systems, networks, etc. described herein as potentially applied to a myriad of other architectures.

Communications in a network environment can be referred to herein as 'messages', 'messaging', 'signaling', 'data', 'content', 'objects', 'requests', 'queries', 'responses', 'replies', etc. which may be inclusive of packets. As referred to herein and in the claims, the term 'packet' may be used in a generic sense to include packets, frames, segments, datagrams, and/or any other generic units that may be used to transmit communications in a network environment. Generally, a packet is a formatted unit of data that can contain control or routing information (e.g., source and destination address, source and destination port, etc.) and data, which is also sometimes referred to as a 'payload', 'data payload', and variations thereof. In some embodiments, control or routing information, management information, or the like can be included in packet fields, such as within header(s) and/or trailer(s) of packets. Internet Protocol (IP) addresses discussed herein and in the claims can include any IP version 4 (IPv4) and/or IP version 6 (IPv6) addresses.

To the extent that embodiments presented herein relate to the storage of data, the embodiments may employ any number of any conventional or other databases, data stores or storage structures (e.g., files, databases, data structures, data or other repositories, etc.) to store information.

Note that in this Specification, references to various features (e.g., elements, structures, nodes, modules, components, engines, logic, steps, operations, functions, characteristics, etc.) included in 'one embodiment', 'example embodiment', 'an embodiment', 'another embodiment', 'certain embodiments', 'some embodiments', 'various embodiments', 'other embodiments', 'alternative embodiment', and the like are intended to mean that any such features are included in one or more embodiments of the present disclosure, but may or may not necessarily be combined in the same embodiments. Note also that a module, engine, client, controller, function, logic or the like as used herein in this Specification, can be inclusive of an executable file comprising instructions that can be understood and processed on a server, computer, processor, machine, compute node, combinations thereof, or the like and may further include library modules loaded during execution, object files, system files, hardware logic, software logic, or any other executable modules.

It is also noted that the operations and steps described with reference to the preceding figures illustrate only some of the possible scenarios that may be executed by one or more entities discussed herein. Some of these operations may be deleted or removed where appropriate, or these steps may be modified or changed considerably without departing from the scope of the presented concepts. In addition, the timing and sequence of these operations may be altered considerably and still achieve the results taught in this disclosure. The preceding operational flows have been offered for purposes of example and discussion. Substantial flexibility is provided by the embodiments in that any suitable arrangements, chronologies, configurations, and timing mechanisms may be provided without departing from the teachings of the discussed concepts.

As used herein, unless expressly stated to the contrary, use of the phrase 'at least one of', 'one or more of', 'and/or', variations thereof, or the like are open-ended expressions that are both conjunctive and disjunctive in operation for any and all possible combination of the associated listed items. For example, each of the expressions 'at least one of X, Y and Z', 'at least one of X, Y or Z', 'one or more of X, Y and Z', 'one or more of X, Y or Z' and 'X, Y and/or Z' can mean any of the following: 1) X, but not Y and not Z; 2) Y, but not X and not Z; 3) Z, but not X and not Y; 4) X and Y, but not Z; 5) X and Z, but not Y; 6) Y and Z, but not X; or 7) X, Y, and Z.

Each example embodiment disclosed herein has been included to present one or more different features. However, all disclosed example embodiments are designed to work together as part of a single larger system or method. This disclosure explicitly envisions compound embodiments that combine multiple previously-discussed features in different example embodiments into a single system or method.

Additionally, unless expressly stated to the contrary, the terms 'first', 'second', 'third', etc., are intended to distinguish the particular nouns they modify (e.g., element, condition, node, module, activity, operation, etc.). Unless expressly stated to the contrary, the use of these terms is not intended to indicate any type of order, rank, importance, temporal sequence, or hierarchy of the modified noun. For example, 'first X' and 'second X' are intended to designate two 'X' elements that are not necessarily limited by any order, rank, importance, temporal sequence, or hierarchy of the two elements. Further as referred to herein, 'at least one of' and 'one or more of' can be represented using the '(s)' nomenclature (e.g., one or more element(s)).

In summary, in some aspects, the techniques described herein relate to a method including: at a control entity configured to communicate with a network: linking a user identifier of a user having an account for call-related information to a hardware identifier of a device to be sent to the user, to produce linked identifiers; loading on the device a limited use token associated with the user that enables only basic call functions on the device, and prevents the device from accessing the account; determining whether the user possesses the device based on the linked identifiers and proximity pairing of the device with a pairing device; and based on determining, either provisioning the device with a full access token that enables full calling functions on the device and grants full access to the account by the device, or not provisioning the device with the full access token to prevent access to the account.

In some aspects, the techniques described herein relate to a method, further including: when the user possesses the device, provisioning the device with the full access token; and when the user does not possess the device, not provisioning the device with the full access token to prevent leakage of account information.

In some aspects, the techniques described herein relate to a method, further including, at the control entity: when the user does not possess the device, remotely disabling the device.

In some aspects, the techniques described herein relate to a method, further including, at the control entity: when the user does not possess the device, providing an alarm that indicates: an attempt has been made to activate the device by an unauthorized user; and an identity of the user.

In some aspects, the techniques described herein relate to a method, wherein determining includes comparing the linked identifiers against information obtained from the device from the proximity pairing.

In some aspects, the techniques described herein relate to a method, wherein determining further includes: upon receiving, from the device, a full provisioning request that carries a reported hardware identifier and a reported user identifier acquired from the pairing device through proximity pairing, evaluating whether the reported hardware identifier and the reported user identifier match the hardware identifier and the user identifier, respectively; when there is a match, confirming that the user possesses the device as paired; and when there is not the match, confirming that the user does not possess the device.

In some aspects, the techniques described herein relate to a method, wherein the account includes: an account identifier; a username; recent call history; a calendar; and billing information.

In some aspects, the techniques described herein relate to a method, further including, at the control entity: before the device has been sent to the user, installing on the device a proximity pairing application that enables proximity pairing based on exchanging, with the pairing device, one of a quick response (QR) code and pairing ultrasound signals.

In some aspects, the techniques described herein relate to a method, wherein the device includes one of an Internet Protocol (IP) phone, a cell phone, and a video conference device.

In some aspects, the techniques described herein relate to a method, wherein the user identifier is associated with an account identifier for the account.

In some aspects, the techniques described herein relate to a method, wherein the basic call functions are limited to receiving phone calls, calling an emergency service, and calling a preloaded information technology (IT) help line for user support for activating the device.

In some aspects, the techniques described herein relate to an apparatus including: a network input/output interface to communicate with a network; and a processor coupled to the network input/output interface and configured to perform: linking a user identifier of a user having an account for call-related information to a hardware identifier of a device to be sent to the user, to produce linked identifiers; loading on the device a limited use token associated with the user that enables only basic call functions on the device, and prevents the device from accessing the account; determining whether the user possesses the device based on the linked identifiers and proximity pairing of the device with a pairing device; and based on determining, either provisioning the device with a full access token that enables full calling functions on the device and grants full access to the account by the device, or not provisioning the device with the full access token to prevent access to the account.

In some aspects, the techniques described herein relate to an apparatus, wherein the processor is further configured to perform: when the user possesses the device, provisioning the device with the full access token; and when the user does not possess the device, not provisioning the device with the full access token to prevent leakage of account information.

In some aspects, the techniques described herein relate to an apparatus, wherein the processor is further configured to perform: when the user does not possess the device, remotely disabling the device.

In some aspects, the techniques described herein relate to an apparatus, wherein the processor is further configured to perform: when the user does not possess the device, providing an alarm that indicates: an attempt has been made to activate the device by an unauthorized user; and an identity of the user.

In some aspects, the techniques described herein relate to an apparatus, wherein the processor is configured to perform determining by comparing the linked identifiers against information obtained from the device from the proximity pairing.

In some aspects, the techniques described herein relate to an apparatus, the processor is further configured to perform determining by: upon receiving, from the device, a full provisioning request that carries a reported hardware identifier and a reported user identifier acquired from the pairing device through proximity pairing, evaluating whether the reported hardware identifier and the reported user identifier match the hardware identifier and the user identifier, respectively; when there is a match, confirming that the user possesses the device as paired; and when there is not the match, confirming that the user does not possess the device.

In some aspects, the techniques described herein relate to a non-transitory computer medium encoded with instructions that, when executed by a processor, cause the processor to perform: linking a user identifier of a user having an account for call-related information to a hardware identifier of a device to be sent to the user, to produce linked identifiers; loading on the device a limited use token associated with the user that enables only basic call functions on the device, and prevents the device from accessing the account; determining whether the user possesses the device based on the linked identifiers and proximity pairing of the device with a pairing device; and based on determining, either provisioning the device with a full access token that enables full calling functions on the device and grants full access to the account by the device, or not provisioning the device with the full access token to prevent access to the account.

In some aspects, the techniques described herein relate to a non-transitory computer medium, further including instructions to cause the processor to perform: when the user possesses the device, provisioning the device with the full access token; and when the user does not possess the device, not provisioning the device with the full access token to prevent leakage of account information.

In some aspects, the techniques described herein relate to a non-transitory computer medium, further including instructions to cause the processor to perform: when the user does not possess the device, remotely disabling the device.

One or more advantages described herein are not meant to suggest that any one of the embodiments described herein necessarily provides all of the described advantages or that all the embodiments of the present disclosure necessarily provide any one of the described advantages. Numerous other changes, substitutions, variations, alterations, and/or modifications may be ascertained to one skilled in the art and it is intended that the present disclosure encompass all such changes, substitutions, variations, alterations, and/or modifications as falling within the scope of the appended claims.

The descriptions of the various embodiments have been presented for purposes of illustration, but are not intended to be exhaustive or limited to the embodiments disclosed. Many modifications and variations will be apparent to those of ordinary skill in the art without departing from the scope and spirit of the described embodiments. The terminology used herein was chosen to best explain the principles of the embodiments, the practical application or technical improvement over technologies found in the marketplace, or to enable others of ordinary skill in the art to understand the embodiments disclosed herein.

What is claimed is:

1. A method comprising:

at a control entity configured to communicate with a network:

linking a user identifier of a user having an account for call-related information to a hardware identifier of a device to be sent to the user, to produce linked identifiers;

loading on the device a limited use token associated with the user that enables only basic call functions on the device, and prevents the device from accessing the account;

determining whether the user possesses the device based on the linked identifiers and proximity pairing of the device with a pairing device; and

based on determining, either provisioning the device with a full access token that enables full calling functions on the device and grants full access to the account by the device, or not provisioning the device with the full access token to prevent access to the account.

2. The method of claim **1**, further comprising:

when the user possesses the device, provisioning the device with the full access token; and

when the user does not possess the device, not provisioning the device with the full access token to prevent leakage of account information.

3. The method of claim **1**, further comprising, at the control entity:

when the user does not possess the device, remotely disabling the device.

4. The method of claim **3**, further comprising, at the control entity:

when the user does not possess the device, providing an alarm that indicates:

an attempt has been made to activate the device by an unauthorized user; and

an identity of the user.

5. The method of claim **1**, wherein determining includes comparing the linked identifiers against information obtained from the device from the proximity pairing.

6. The method of claim **5**, wherein determining further includes:

upon receiving, from the device, a full provisioning request that carries a reported hardware identifier and a reported user identifier acquired from the pairing device through the proximity pairing, evaluating whether the reported hardware identifier and the reported user identifier match the hardware identifier and the user identifier, respectively;

when there is a match, confirming that the user possesses the device as paired; and

when there is not the match, confirming that the user does not possess the device.

7. The method of claim **1**, wherein the account includes: an account identifier; a username; recent call history; a calendar; and billing information.

8. The method of claim **1**, further comprising, at the control entity:

before the device has been sent to the user, installing on the device a proximity pairing application that enables the proximity pairing based on exchanging, with the pairing device, one of a quick response (QR) code and pairing ultrasound signals.

9. The method of claim **1**, wherein the device includes one of an Internet Protocol (IP) phone, a cell phone, and a video conference device.

10. The method of claim **1**, wherein the user identifier is associated with an account identifier for the account.

11. The method of claim **1**, wherein the basic call functions are limited to receiving phone calls, calling an emergency service, and calling a preloaded information technology (IT) help line for user support for activating the device.

12. An apparatus comprising:

a network input/output interface to communicate with a network; and

a processor coupled to the network input/output interface and configured to perform:

linking a user identifier of a user having an account for call-related information to a hardware identifier of a device to be sent to the user, to produce linked identifiers;

loading on the device a limited use token associated with the user that enables only basic call functions on the device, and prevents the device from accessing the account;

determining whether the user possesses the device based on the linked identifiers and proximity pairing of the device with a pairing device; and

based on determining, either provisioning the device with a full access token that enables full calling functions on the device and grants full access to the account by the device, or not provisioning the device with the full access token to prevent access to the account.

13. The apparatus of claim **12**, wherein the processor is further configured to perform:

when the user possesses the device, provisioning the device with the full access token; and

when the user does not possess the device, not provisioning the device with the full access token to prevent leakage of account information.

14. The apparatus of claim **12**, wherein the processor is further configured to perform:

when the user does not possess the device, remotely disabling the device.

15. The apparatus of claim **14**, wherein the processor is further configured to perform:

when the user does not possess the device, providing an alarm that indicates:

an attempt has been made to activate the device by an unauthorized user; and

an identity of the user.

16. The apparatus of claim **12**, wherein the processor is configured to perform determining by comparing the linked identifiers against information obtained from the device from the proximity pairing.

17. The apparatus of claim **16**, the processor is further configured to perform determining by:

upon receiving, from the device, a full provisioning request that carries a reported hardware identifier and a reported user identifier acquired from the pairing device through the proximity pairing, evaluating whether the reported hardware identifier and the reported user identifier match the hardware identifier and the user identifier, respectively;

when there is a match, confirming that the user possesses the device as paired; and

when there is not the match, confirming that the user does not possess the device.

18. A non-transitory computer medium encoded with instructions that, when executed by a processor, cause the processor to perform:

linking a user identifier of a user having an account for call-related information to a hardware identifier of a device to be sent to the user, to produce linked identifiers;

loading on the device a limited use token associated with the user that enables only basic call functions on the device, and prevents the device from accessing the account;

determining whether the user possesses the device based on the linked identifiers and proximity pairing of the device with a pairing device; and

based on determining, either provisioning the device with a full access token that enables full calling functions on the device and grants full access to the account by the device, or not provisioning the device with the full access token to prevent access to the account.

19. The non-transitory computer medium of claim **18**, further comprising instructions to cause the processor to perform:

when the user possesses the device, provisioning the device with the full access token; and

when the user does not possess the device, not provisioning the device with the full access token to prevent leakage of account information.

20. The non-transitory computer medium of claim **18**, further comprising instructions to cause the processor to perform:

when the user does not possess the device, remotely disabling the device.

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